

MADHAVAN S

+91 8300838382 | madhavankky5@gmail.com | ee24ms003@iitdh.ac.in

 [madhavan-s](#)

Dharwad, India

OBJECTIVE

Seeking opportunities to develop innovative solutions in robotics and autonomous systems and more, using communications, signal processing, machine learning (and more).

EDUCATION

- **IIT Dharwad** 2024 - Present
M.Tech (R) in Communications, Signal Processing and Machine Learning
Dharwad, India
 - CGPA: 7.82/10.00 (Till IV semester)
- **SASTRA Deemed University** 2020 - 2024
B.Tech in Electronics and Communications
Tamil Nadu, India
 - CGPA: 8.45/10.00

PROJECTS

- **Hybrid Data-Driven Search for Multi-Agent Autonomous Exploration** 2024 - Present
Masters Thesis, Indian Institute of Technology Dharwad
 - Working on hybrid data-driven search framework for autonomous exploration using multi-armed bandits (MABs).
 - Identifying interesting regions using MABs which are further broken into cells and are classified into 'keep'/'reject'.
 - Focusing on efficient exploration, agent coordination and redundancy avoidance in partially observable environments.
- **Stock Price Estimation** 2025
Course Project(Detection and Estimation), IIT Dharwad
 - Applied maximum likelihood estimation (MLE), Bayesian inference, to predict short-term stock prices using historical log returns.
 - Developed a hybrid model integrating MLE and Bayesian estimates with adaptive inverse variance weighting for robust and accurate stock price forecasting.
 - Evaluated models with MAE, RMSE, MAPE, and R^2 , with the hybrid model achieving the lowest MAE (3.2548), RMSE (3.2548), MAPE (1.8877), and highest R^2 (0.4159).
- **Paper-based Microwave Sensor for Detection of Volatile Organic Compounds (VOCs)** 2023
B.Tech Mini Project, SASTRA Deemed University
 - Designed and fabricated a cost-effective, eco-friendly microstrip patch antenna using Whatman filter paper to detect volatile organic compounds (VOCs).
 - Utilized Keysight ADS software to simulate and optimize antenna performance with split-ring resonator structures for enhanced VOC sensing.
 - Achieved successful VOC detection by analyzing resonance frequency shifts with a network analyzer, contributing to advancements in environmental monitoring.
- **Raga Identification from Arohana and Avarohana Patterns using PNCC and Clusters** 2024
B.Tech Major Project, SASTRA Deemed University
 - Developed a system to identify Indian classical ragas from note patterns using power normalized cepstral coefficients (PNCC).
 - Collected and pre-processed audio data to extract PNCC features.
 - Implemented K-means clustering for classification.
 - Trained a model in MATLAB, achieving 89% accuracy.

EXPERIENCE

• TCS Research and Innovation

Jan 2026 – Jun 2026

Robotics Intern

- Built a real-time bimanual teleoperation of Piper arms using Meta Quest 3, used in the **2nd WBCD Challenge, IEEE ICRA 2026**.
- Developed a low-latency stereoscopic vision pipeline streaming live ZED Mini stereo footage to the Quest 3 headset.
- Trained and evaluated an ACT (Action Chunking with Transformers) policy on the pick dual bottles task in simulation using the AgileX Piper embodiment, successfully completing dual-bottle pick-and-place to target positions.

• ONGC

August 2022

Summer Intern

- Analysed the electronics and communication systems used in INFOCOM at ONGC-Cauvery Asset, including SCADA, networking, exchange, DISHA.

PUBLICATIONS

C=CONFERENCE

- [C.1] **Madhavan S** and G. K. Kamath M, "*Hierarchical Bandit Monitoring with Nested Grid Refinement Using Mobile Sensors and Charging Stations*", to be submitted to American Control Conference (ACC).
- [C.2] **Madhavan S** and G. K. Kamath M, "*Sparsity-Aware and Spatially-Guided Bandit Monitoring with Mobile Sensors and Charging Stations*", to be submitted to American Control Conference (ACC).

SKILLS

- **Domain Expertise :** Machine Learning, Reinforcement Learning, Signal Processing, Optimization-based Decision Making, Autonomous Systems, Multi-Armed Bandits
- **Programming Languages:** Python, MATLAB, C/C++
- **Python Libraries:** NumPy, Gurobi, Pandas, SciPy
- **Algorithms and Techniques:** Estimation Algorithms (MLE, Bayesian Inference), Detection Algorithms, Clustering (K-means), Linear & Quadratic Programming, Path Planning, Multi-Agent Coordination
- **Tools and Simulation:** ROS2, Gazebo, ADS

ADDITIONAL INFORMATION

◦ RELEVANT COURSES:

1. **Communication:** Digital communication, analog communication, optical communication, wireless communication, information theory and coding, detection and estimation theory.
2. **Signal Processing:** Signals and systems, digital signal processing, probability model and its applications, linear algebra, optimization theory and algorithms.
3. **RF System:** RF Systems, Antenna and wave propagation.

◦ PROFESSIONAL DEVELOPMENT:

Matlab programming (Mathworks), Python programming (Udemy).

◦ WORKSHOPS ATTENDED:

1. Introduction to ADS, EMPro (SASTRA University 2023).
2. Workshop on Proteus Simulation .

◦ LANGUAGES: Tamil, English, Hindi, Malayalam (beginner).

REFERENCES

1. Prof Gopal Krishna Kamath M

Assistant Professor, Electrical, Electronics and Communication Engineering
IIT Dharwad
Email: gopal.kamath@iitdh.ac.in
Phone: +91-9952046432
Relationship: Supervisor at IIT Dharwad

2. Dr. Kaushik Das

Senior Scientist, TCS Research
Email: kaushik.da@tcs.com
Phone: +91-9980239777
Relationship: Supervisor/Mentor at TCS Research and Innovation

3. Prof. B N Bharath

Associate Professor, Electrical, Electronics and Communication Engineering
IIT Dharwad
Email: bharathbn@iitdh.ac.in
Phone: +91-836-2309664
Relationship: Course Instructor at IIT Dharwad